UNIT 4

NETWORK LAYER

Virtual – circuit network

- Virtual – circuit network is a category of packet switching network, where a virtual path is established between the source and the destination systems for data communication to occur.
- This path appears to the user as if it is a dedicated physical path, but actually is a logical circuit allocated from a managed pool of circuit resources as per traffic requirements.
- The network resources forming parts of this path can be shared by other communications, however, is not visible to this user.

Features of Virtual – Circuit Network

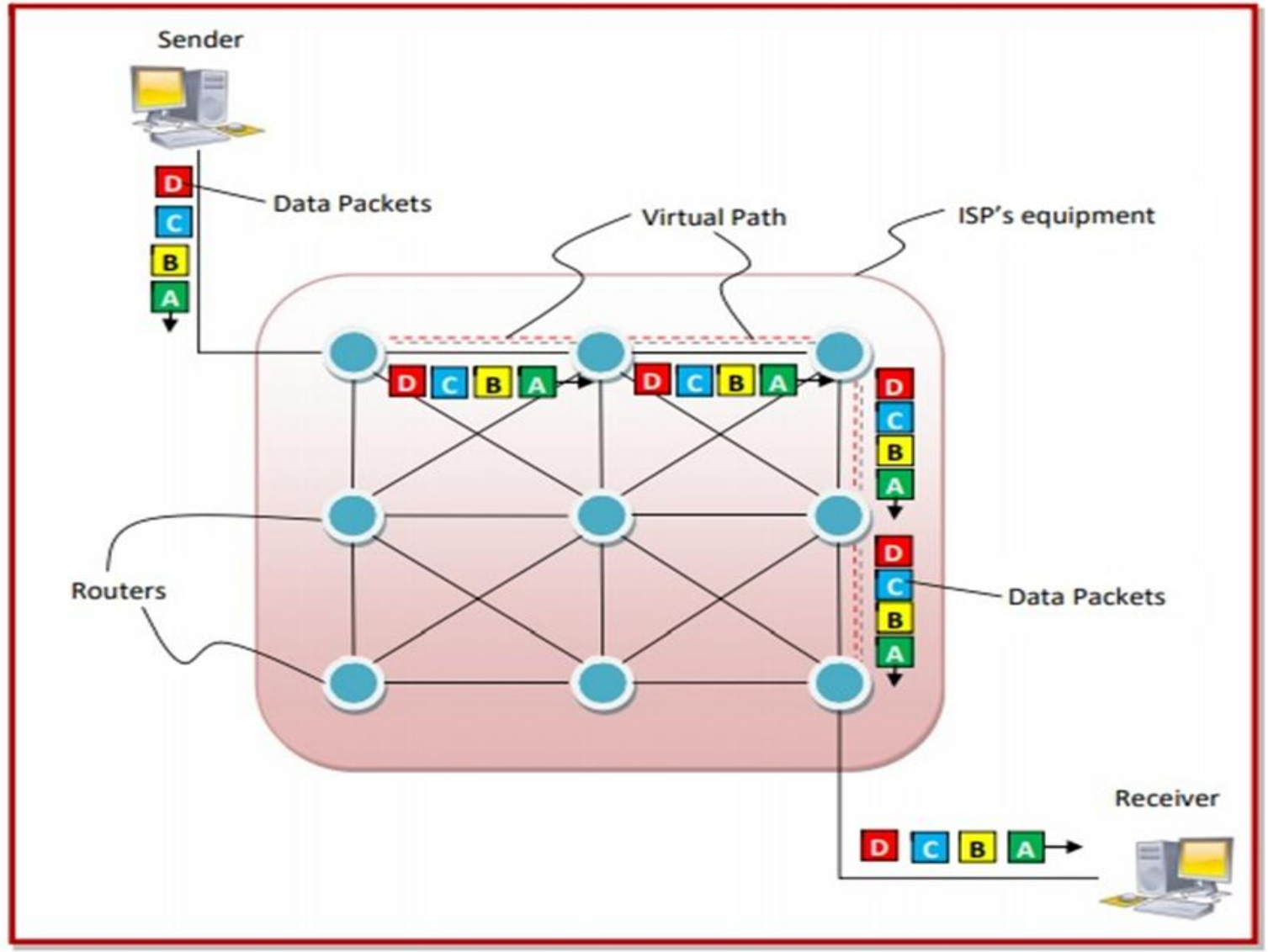
- Virtual – circuit switching is done at the network layer of the communication system.
- In these networks, the path between the source and the destination nodes that is followed by first data packet gets allocated. All other data packets transmitted between them will use the same path.
- All the resources like buffers and bandwidth get reserved before the transmission, and all data packets consume same resources.
- Since all packets follow the same path, a common header and routing information is used by them.
- They provide greater reliability and less complexity owing to fixed paths and fixed resources.

Phases of Virtual - Circuit Transmission

- **Set up Phase** − In this phase, a virtual circuit or a route is established from the source to the destination through number of switches. The source and destination use global addresses using which the switches make routing table entries.

- **Data Transfer** − Once the virtual circuit is set up, all packets follow the route established during the set up phase adhering to the routing tables.

- **Teardown Phase** − When data transfer is complete, the source sends a teardown request. The destination responds using a teardown confirmation. The switches flush their routing table entries, thus relinquishing the circuit.


Sender
Data Packets
Virtual Path
ISP's equipment
Routers
Data Packets
Receiver

Congestion Control in Virtual Circuit

Once the congestion is detected in virtual circuit network, closed-loop techniques is used. There are different approaches in this technique:

- **No new connection -** No new connections are established when the congestion is detected. This approach is used in telephone networks where no new calls are established when the exchange is overloaded.
- **Participation of congested router invalid -** Another approach to control congestion is allow all new connections but route these new connections in such a way that congested router is not part of this route.
- **Negotiation -** To negotiate different parameters between sender and receiver of the network, when the connection is established. During the set up time, host specifies the shape and volume of the traffic, quality of service and other parameters.

Advantages of Virtual Circuit:

1. Packets are delivered to the receiver in the same order sent by the sender.
2. Virtual circuit is a reliable network circuit.
3. There is no need for overhead in each packet.
4. Single global packet overhead is used in virtual circuit.

Disadvantages of Virtual Circuit:

1. Virtual circuit is costly to implement.
2. It provides only connection-oriented service.
3. Always a new connection set up is required for transmission.

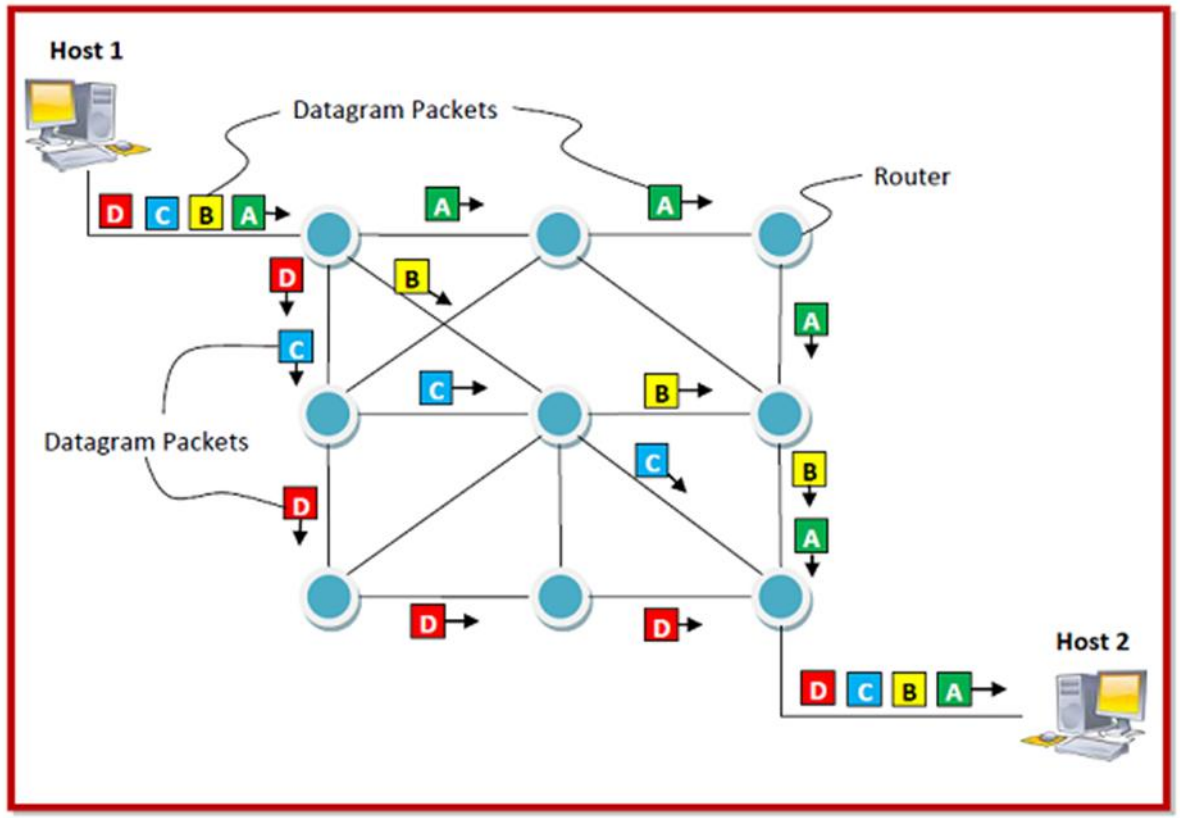
Datagram Network

- In a connectionless communication systems, datagram refers to the smallest unit via which data is transmitted.
- Datagrams are data packets which contain adequate header information so that they can be individually routed by all intermediate network switching devices to the destination.
- These networks are called datagram networks since communication occurs via datagrams. They exist in packet switching networks.

Features of Datagram Networks

- Datagram switching is done at the network layer of the communication system.
- In datagram networks, each data packet or datagram is routed independently from the source to the destination even if they belong to the same message. The network treats the packet as if it exists alone.
- Since the datagrams are treated as independent units, no dedicated path is fixed for data transfer. Each datagram is routed by the intermediate routers using dynamically changing routing tables. So two successive packets from the source may follow completely separate routes to reach destination.
- In these networks, no prior resource allocation is done for the individual packets. This implies that no resources like buffers, processors, bandwidth, etc. are reserved before the communication commences.
- In datagram networks, resources are allocated on demand on a First−Come First−Serve (FCFS) basis. When a packet arrives at a router, the packet must wait if there are other packets being processed, irrespective of its source or destination.
- Datagram communication is generally guided by User Datagram Protocol or UDP

The following diagram shows datagram packets being send by host H1 to host H2. The four datagram packets labelled as A, B, C and D, all belonging to same message are being routed separately via separate routes. The packets in the message arrives in the destination out of order. It is the responsibility of H2 to reorder the packets in order to retrieve the original message.

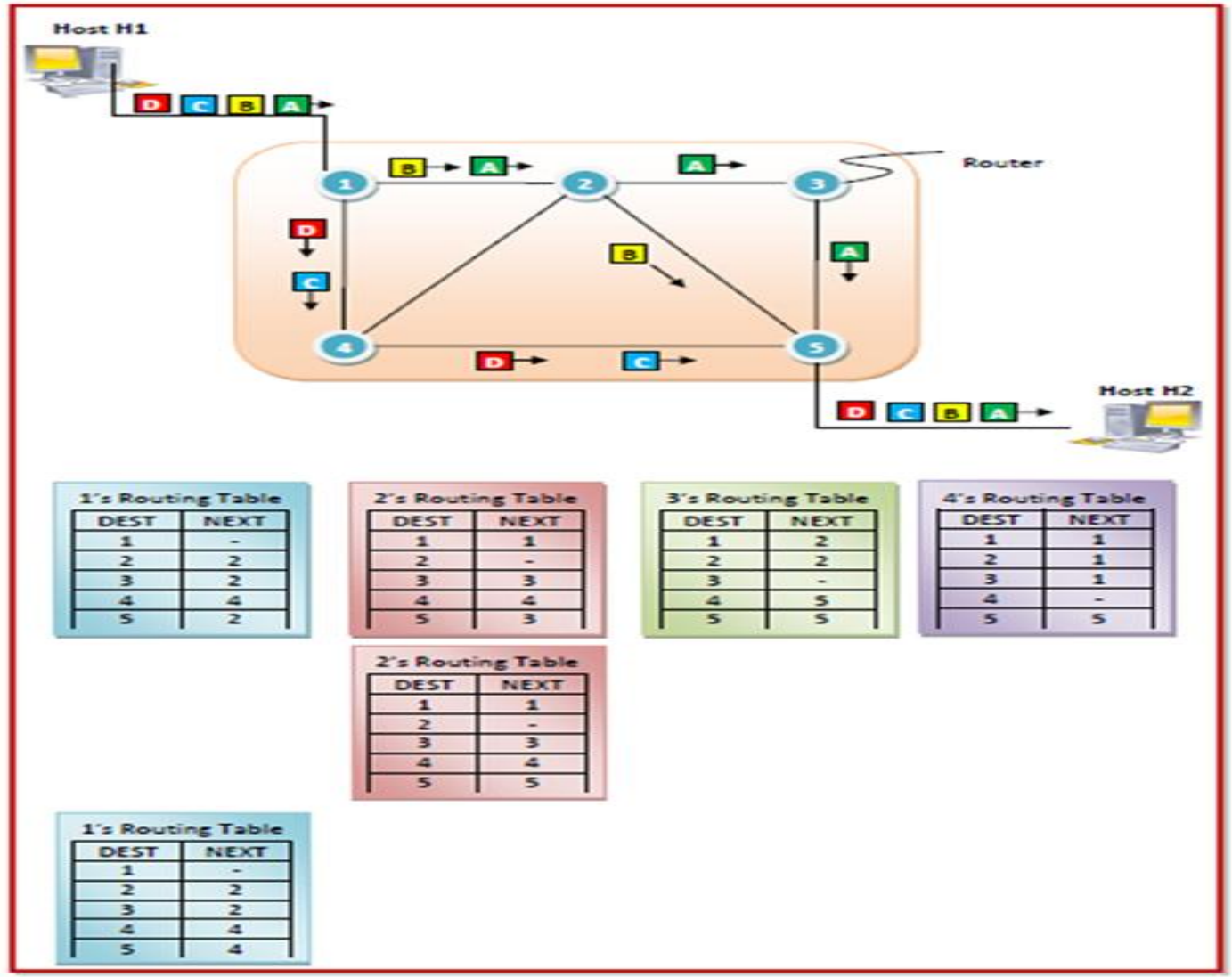


Routing Principle

- In datagram networks, each data packet is routed independently from the source to the destination even if they belong to the same message.
- No prior resource or channel allocation is done for the individual packets.
- As the datagrams are treated as independent units, no dedicated path is fixed for data transfer.
- Each datagram is routed by the intermediate routers using dynamically changing routing tables.
- So two successive packets from the source may follow completely separate routes to reach destination.
- Resources are allocated on demand on a First-Come First-Serve (FCFS) basis.
- When a packet arrives at a router, the packet must wait if there are other packets being processed, irrespective of its source or destination.

Example of Routing in a Datagram Network

- The diagram shows a message being send by host H1 to host H2. The Internet Service Provider (ISP) comprises of five routers that are numbered from 1 to 5. H1 is connected to router 1 while H2 is connected to router 5.
- Suppose that the message is of such a size that it has to be broken into 4 packets. The packets are labelled as A, B, C and D.
- Each of the five routers maintains a routing table that has two columns, DEST storing the destination node and NEXT storing next node. To reach the DEST node the packet is routed via the NEXT node. The routing tables are dynamic in nature that are refreshed time to time depending upon the network conditions.



1's Routing Table

DEST	NEXT
1	-
2	2
3	2
4	4
5	2

2's Routing Table

DEST	NEXT
1	1
2	-
3	3
4	4
5	3

3's Routing Table

DEST	NEXT
1	2
2	2
3	-
4	5
5	5

4's Routing Table

DEST	NEXT
1	1
2	1
3	1
4	-
5	5

2's Routing Table

DEST	NEXT
1	1
2	-
3	3
4	4
5	5

1's Routing Table

DEST	NEXT
1	-
2	2
3	2
4	4
5	4

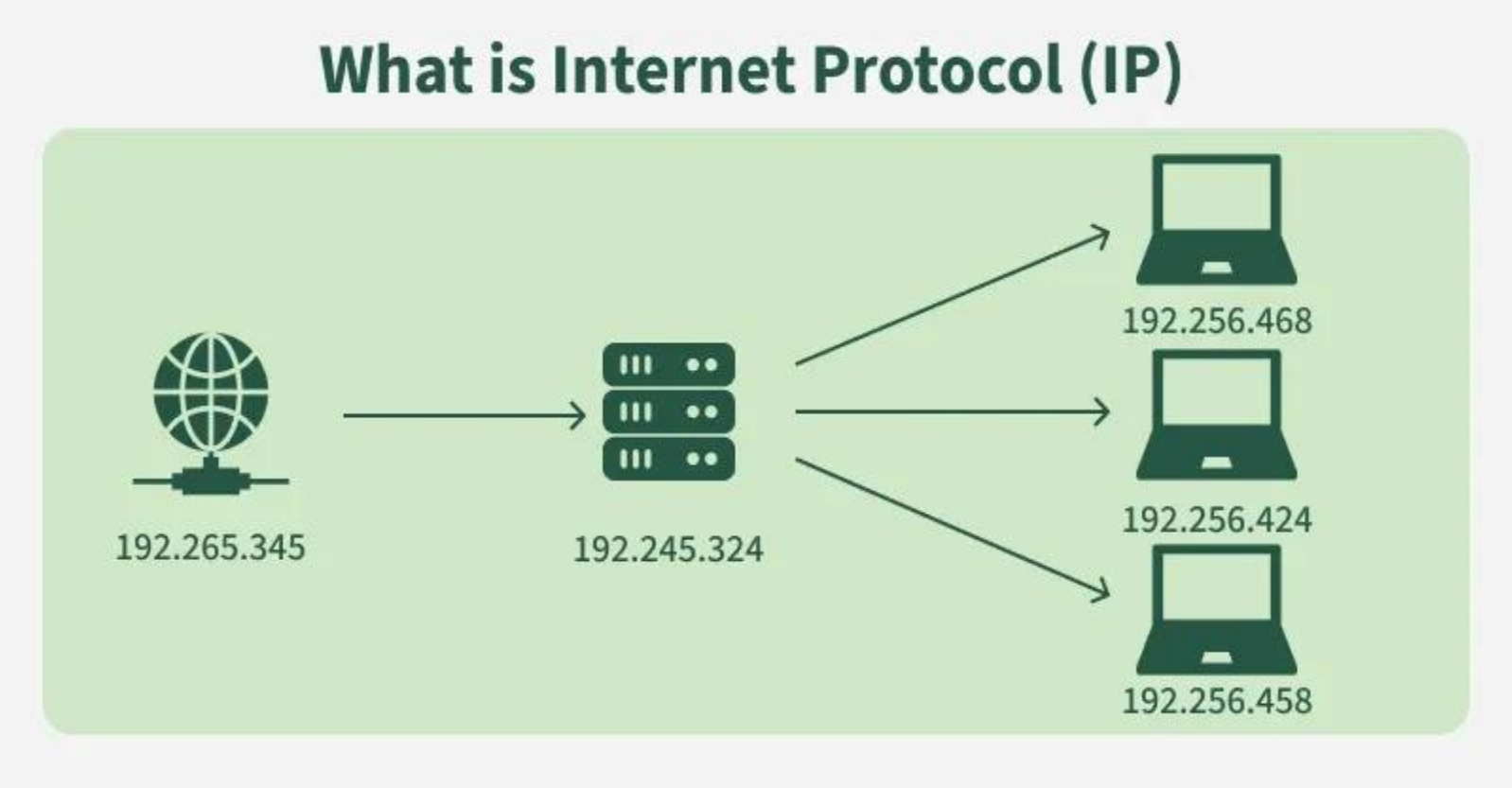
Datagram Approach	Virtual Circuit
There is no dedicated transmission path.	There is also no dedicated transmission path.
There is a network responsible for individual packets.	There is a network responsible for packet sequences.
The route is established for each table.	The route is established for the entire conversation.
It is suitable for a few packets in data communication.	It is suitable for longer data transmission.
There is no connection established.	Connection is established.
Datagram approaches are not dependable because the connectionless property creates data packets that reach the destination. It is different from the order in which they were transmitted.	It provides a dependable communication function to provide that each group appears correctly and the sets remain in the original order in which they were transmitted.
It is frequently used in the IP network which can be used for data service such as the internet.	It can be used by the ATM (Asynchronous Transfer Mode) Network, which can be used for mobile calls.
The packets arrive at their intended destination in a multiple order in which they were transmitted.	The packets continually reach their destined destination in the similar order in which they were transmitted.

Internet Protocol (IP)

- The Internet Protocol (IP) is a set of rules that allows computers and other devices to communicate over the Internet.
- It ensures that information sent from one device reaches the correct destination by using a unique set of numbers known as IP addresses.

Continued…

Whether you're browsing websites, sending emails, or watching videos, the Internet Protocol helps manage how information travels, making sure everything works smoothly and efficiently.



Why is Internet Protocol Important?

- The Internet Protocol is a fundamental component of the Internet and computer networks, responsible for delivering packets of data from the source host to the destination host based on their IP addresses. It ensures that packets of data get to the right destination from the source device.
- Each device connected to a network is assigned an IP address, which serves as a "**home address**" for the device, enabling other devices to locate and send messages to it and providing the addressing and routing mechanisms the devices require for their communications.
- IP works in conjunction with the Internet Control Message protocol(ICMP) to send error messages and operational information (e.g., destination unreachable, echo requests), which inform senders about issues in packet delivery.

Primary Terminologies



IP Address



Packet



Router



IPv4,IPv6

These terms form the foundation for understanding how devices communicate and exchange data efficiently across a network.

- **IP Address: IP address** is referred as a number sticker given to each device that belongs to the network which utilizes Internet Protocol to communicate. It serves two main purposes: host or network interface recognition, identifier or location addressing.
- **Packet:** A packet is a parcel of data that is switched between an origin and a destination via the Internet or some other network that is based on the packet switching mechanism. This component mainly has a header and a payload.
- **Router:** A router is a network device that (serves as) a forwarding point for data packets between computer networks. Routers carry out the traffic routing functions through the Internet.

Continued…

- **IPv4:** Internet Protocol version 4 (IPv4) is a protocol that tends to provide connectivity between the desktop computers online and that application layer is the wider application one that covers most of the Internet communications nowadays
- **IPv6:** Internet Protocol version 6 (IPv6), the final iteration in the series of the upgrades to the Internet Protocol, is used for the purpose of identifying, locating and routing of various computers so that traffic through Internet can be transferred properly.

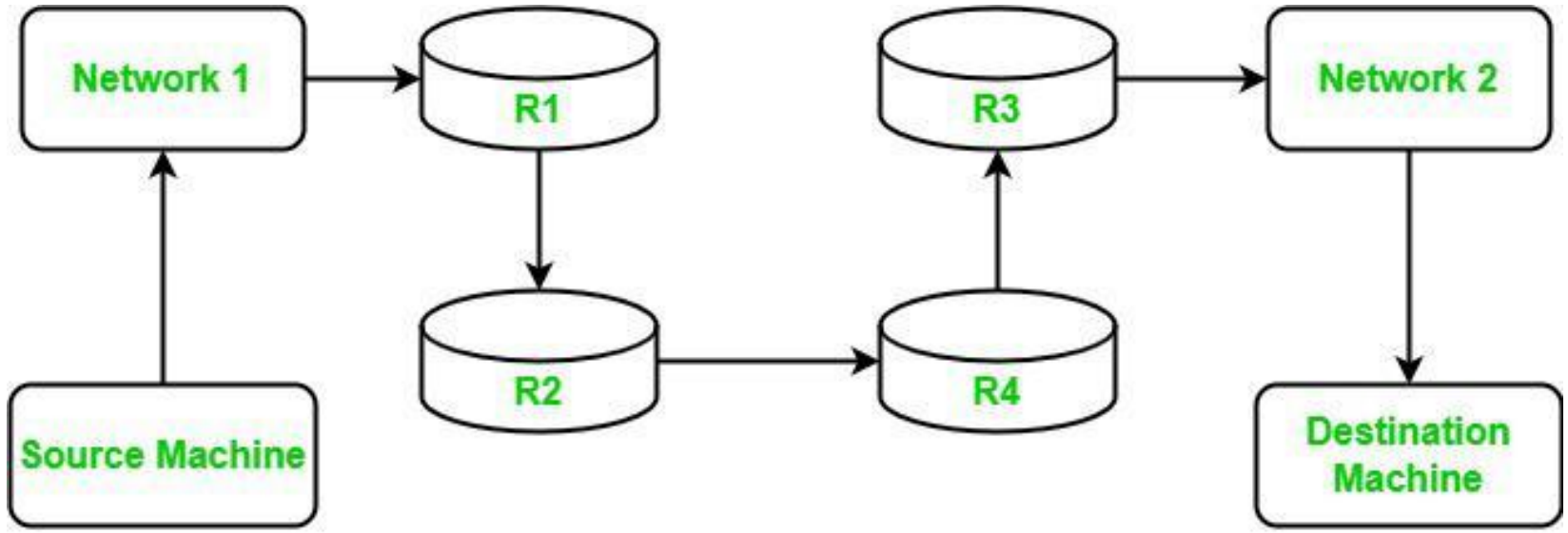
What is an IP Packet?

An IP packets is the basic unit of data transmission in an IP network. It consists of a header and a payload:

- **Header**: The header contains essential control information, such as the source and destination IP addresses, that helps routers determine where to send the packet.
- **Payload**: The payload contains the actual data being transmitted. Once the packet reaches its destination, the data in the payload is delivered to the appropriate application or protocol

How Does the IP Routing Work?

- IP routing is a procedure of routing the information from the source to the recipient to enable its direction to the final destination.
- As we also know that the data is broken into several pieces, and then each piece will almost definitely go through several routers until these data ends up in the final destination.
- The path that the set of data package follows is determined by the routing algorithm.
- Besides the size of packet and the header length, the design of the routing algorithm takes into account other factors related to determining the most suitable route for the data from the sender to the destination.
- At the arrival of the datagram at some router, then source address and destination address fields are employed with a routing table to decide the subsequent hop's address.


Network 1
R1
R3
Network 2
R2
R4
Source Machine
Destination Machine

IP Routing

IPv4

- **IP** stands for **Internet Protocol version v4** stands for **Version Four** (IPv4), is the most widely used system for identifying devices on a network.
- It uses a set of four numbers, separated by periods (like 192.168.0.1), to give each device a unique address.
- This address helps data find its way from one device to another over the internet.

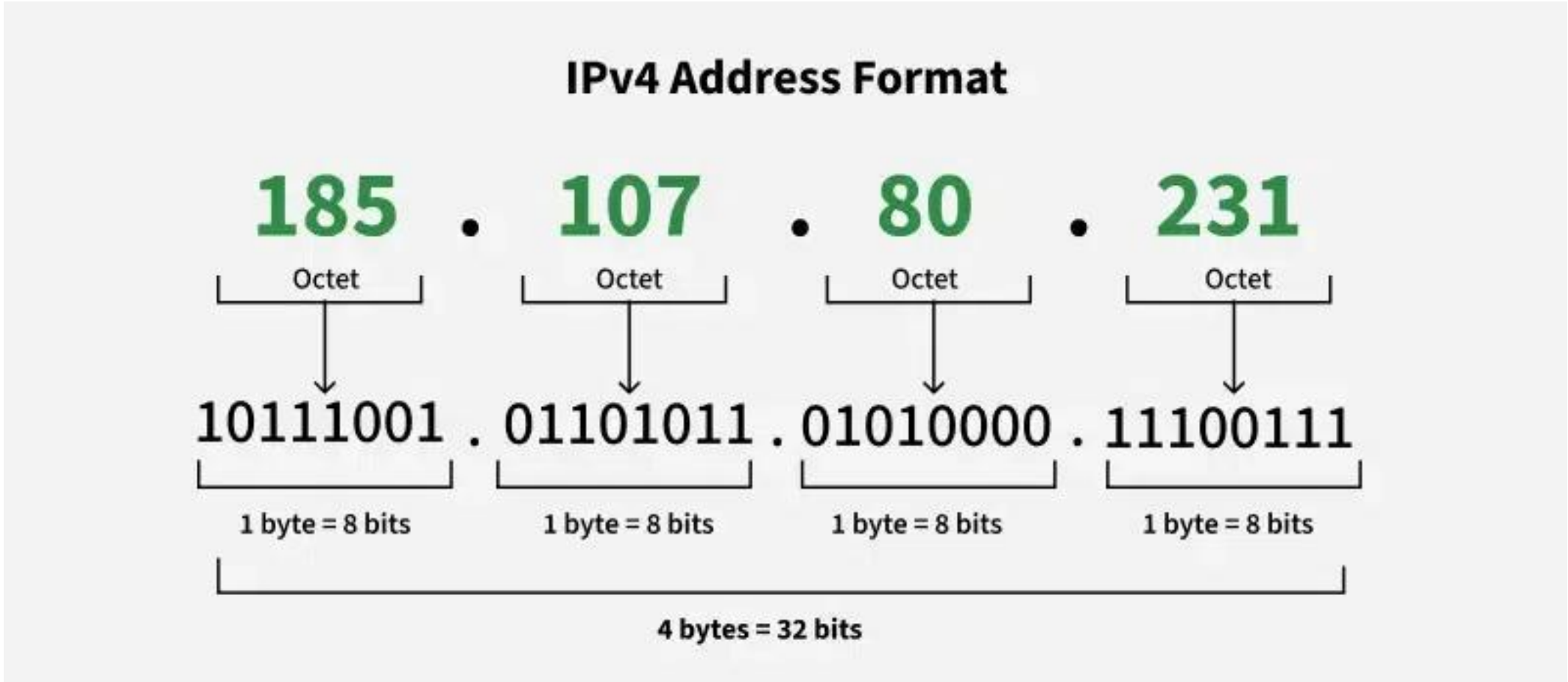
IPv4 Addressing

- An IPv4 address consists of series of four eight-bit binary numbers which are separated by decimal point.
- Although any numbering system can be used to represent a unique 32- bit number, most commonly you see IP address expressed in dot decimal notation.

Site	Dot-decimal	Binary
Twitter.com	104.244.42.129	01101000.11110100.00101010.10000001
Reddit.com	151.101.65.140	10010111.01100101.01000001.10001100
Linkedin.com	108.174.10.10	01101100.10101110.00001010.00001010

IPv4 Address Format

- An IPv4 address consists of 32 bit (binary digit), grouped into four section of known as **octets or bytes.**
- **Each octet** has 8 bits and this bits can be represented only in 0 or 1 form, and when they grouped together, they form a binary number.
- Since each octet has 8 bits, it can represent 256 numbers ranging from o to 255.
- These four octets are represented as decimal numbers, separated by periods known as **dotted decimal notation**.
- For example IPv4 address **185.107.80.231** consists of four octets.


IPv4 Address Format
185 . 107 . 80 . 231
Octet
Octet
Octet
Octet
10111001 . 01101011 . 01010000 . 11100111
1 byte = 8 bits
1 byte = 8 bits
1 byte = 8 bits
1 byte = 8 bits
4 bytes = 32 bits

Example 19.1

Change the following IPv4 addresses from binary notation to dotted-decimal notation.

a. 10000001 00001011 00001011 11101111

b. 11000001 10000011 00011011 11111111

Solution

We replace each group of 8 bits with its equivalent decimal number (see Appendix B) and add dots for separation.

a. 129.11.11.239

b. 193.131.27.255

Parts of IPv4

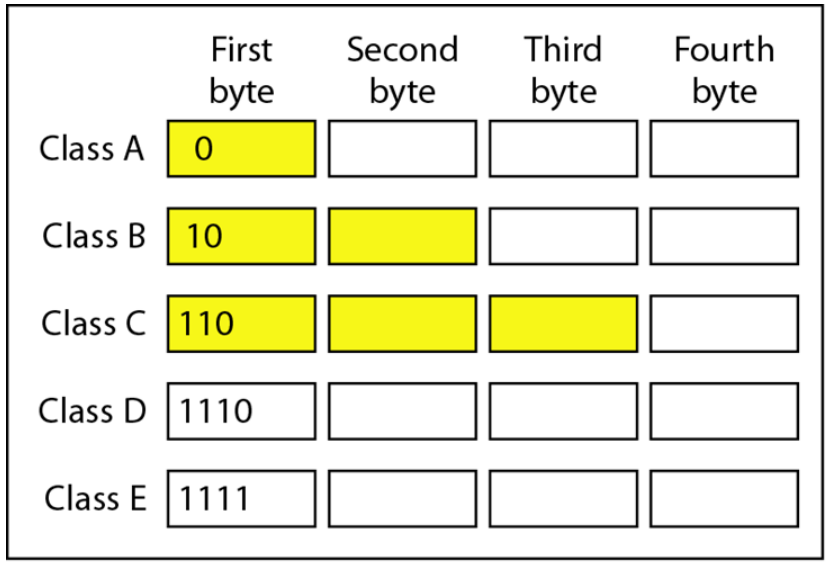
IPv4 addresses consist of three parts:

- **Network Part**: The network part indicates the distinctive variety that's appointed to the network. The network part conjointly identifies the category of the network that's assigned.
- **Host Part**: The host part uniquely identifies the machine on your network. This part of the IPv4 address is assigned to every host.
  For each host on the network, the network part is the same, however, the host half must vary.
- **Subnet Number**: This is the non obligatory part of IPv4. Local networks that have massive numbers of hosts are divided into subnets and subnet numbers are appointed to that.

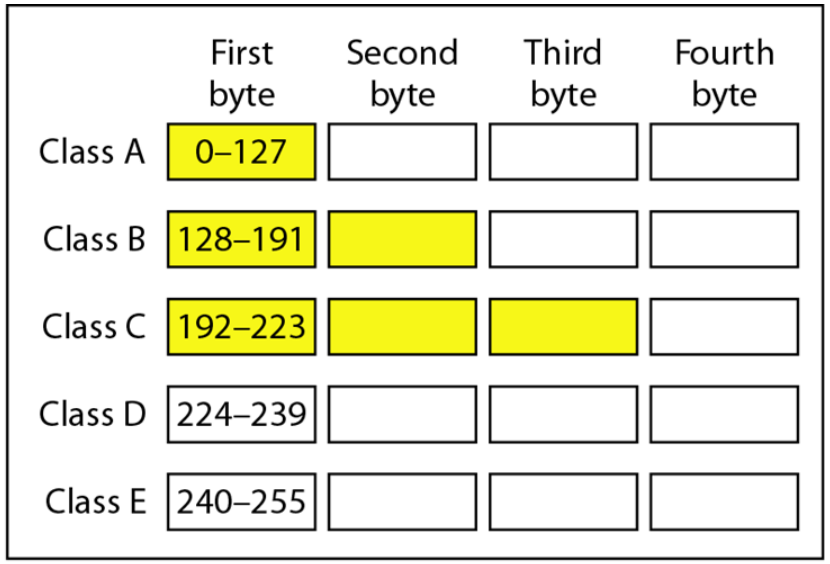
Note

In classful addressing, the address space is divided into five classes: A, B, C, D, and E.

Figure 19.2 *Finding the classes in binary and dotted-decimal notation*



a. Binary notation



b. Dotted-decimal notation

Example 19.4

Find the class of each address.

a. 00000001 00001011 00001011 11101111

b. 11000001 10000011 00011011 11111111

c. 14.23.120.8

d. 252.5.15.111

Solution

a. ***The first bit is 0. This is a class A address.***

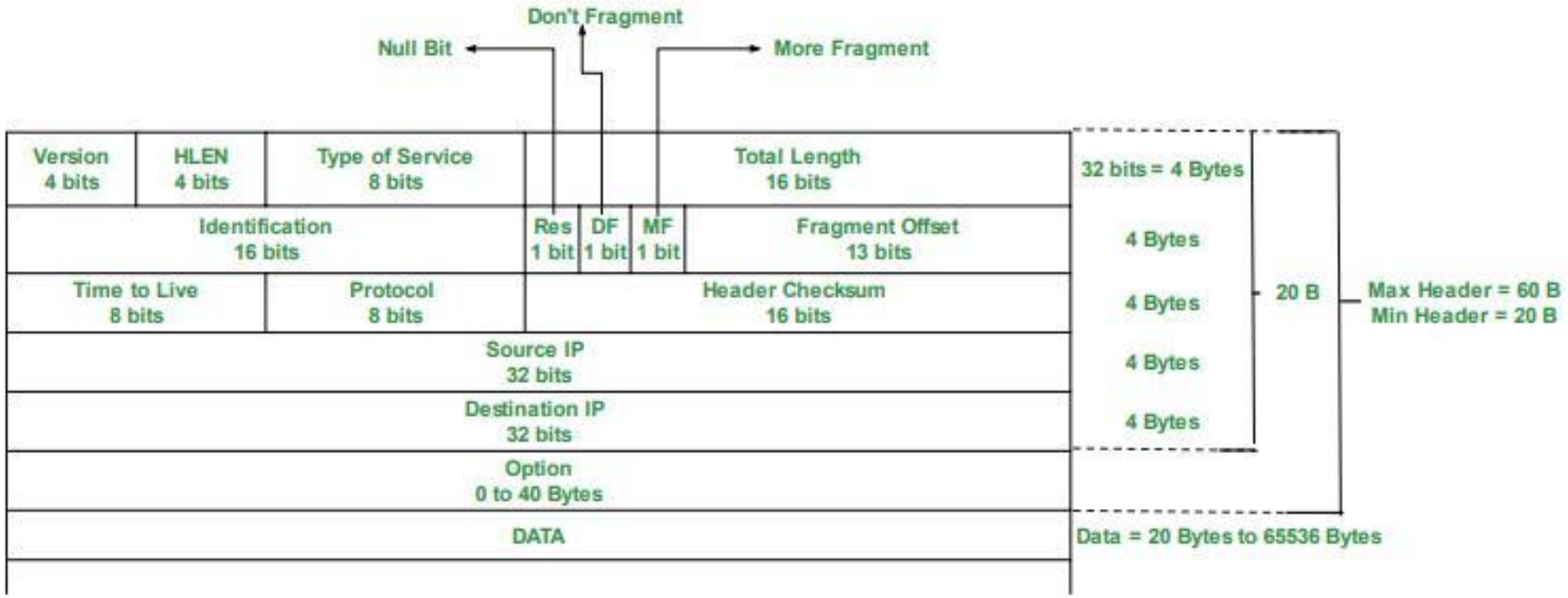
b. ***The first 2 bits are 1; the third bit is 0. This is a class C address.***

c. ***The first byte is 14; the class is A.***

d. ***The first byte is 252; the class is E.***

IPv4 Datagram Header

These fields together ensure the proper delivery, routing, fragmentation, and reassembly of IP packets across the network.



•**VERSION*:*** Version of the IP protocol (4 bits), which is 4 for IPv4

•**HLEN:** IP header length (4 bits), which is the number of 32 bit words in the header. The minimum value for this field is 5 and the maximum is 15.

•**Type of service:** Low Delay, High Throughput, Reliability (8 bits)

•**Total Length:** Length of header + Data (16 bits), which has a minimum value 20 bytes and the maximum is 65,535 bytes.

•**Identification*:*** Unique Packet Id for identifying the group of fragments of a single IP datagram (16 bits)

•**Flags*:*** 3 flags of 1 bit each : reserved bit (must be zero), do not fragment flag, more fragments flag (same order)

•**Fragment Offset:** Represents the number of Data Bytes ahead of the particular fragment in the particular Datagram. Specified in terms of number of 8 bytes, which has the maximum value of 65,528 bytes.

•**Time to live*:*** Datagram’s lifetime (8 bits), It prevents the datagram to loop through the network by restricting the number of Hops taken by a Packet before delivering to the Destination.

•**Protocol:** Name of the protocol to which the data is to be passed (8 bits)

•**Header Checksum*:*** 16 bits header checksum for checking errors in the datagram header

•**Source IP address*:*** 32 bits IP address of the sender

•**Destination IP address*:*** 32 bits IP address of the receiver

•**Option:** Optional information such as source route, record route. Used by the Network administrator to check whether a path is working or not.

Advantages of IPv4

- IPv4 security permits encryption to keep up privacy and security.
- IPV4 network allocation is significant and presently has quite 85000 practical routers.
- It becomes easy to attach multiple devices across an outsized network while not NAT.
- This is a model of communication so provides quality service also as economical knowledge transfer.
- IPV4 addresses are redefined and permit flawless encoding.
- IPv4 has high System Management prices and it's labor-intensive, complex, slow & prone to errors.
- Routing is scalable and economical as a result of addressing its collective more effectively.
- Data communication across the network becomes a lot of specific in multicast organizations.

Limitations of IPv4

- IP relies on network layer addresses to identify end-points on the network, and each network has a unique IP address.
- The world's supply of unique IP addresses is dwindling, and they might eventually run out theoretically.
- If there are multiple hosts, we need the IP addresses of the next class.
- Complex host and routing configuration, non-hierarchical addressing, difficult to re-numbering addresses, large routing tables, non-trivial implementations in providing security, QoS (Quality of service), mobility, and multi-homing, multicasting, etc. are the big limitations of IPv4 so that's why IPv6 came into the picture.

IPv6

- The Internet Protocol version 6, or IPv6, is the latest version of the Internet Protocol (IP), which is the system used for identifying and locating computers on the Internet.
- IPv6 was developed by the Internet Engineering Task Force (IETF) .
- IPv6 is a 128-bit address having an address space of 2^{128}, which is way bigger than IPv4.
- IPv6 uses a Hexa-Decimal format separated by a colon (:)

Components in IPv6 Address Format

- There are 8 groups and each group represents 2 Bytes (16-bits).
- Each Hex-Digit is of 4 bits (1 nibble)
- Delimiter used - colon (:)
-



Need For IPv6

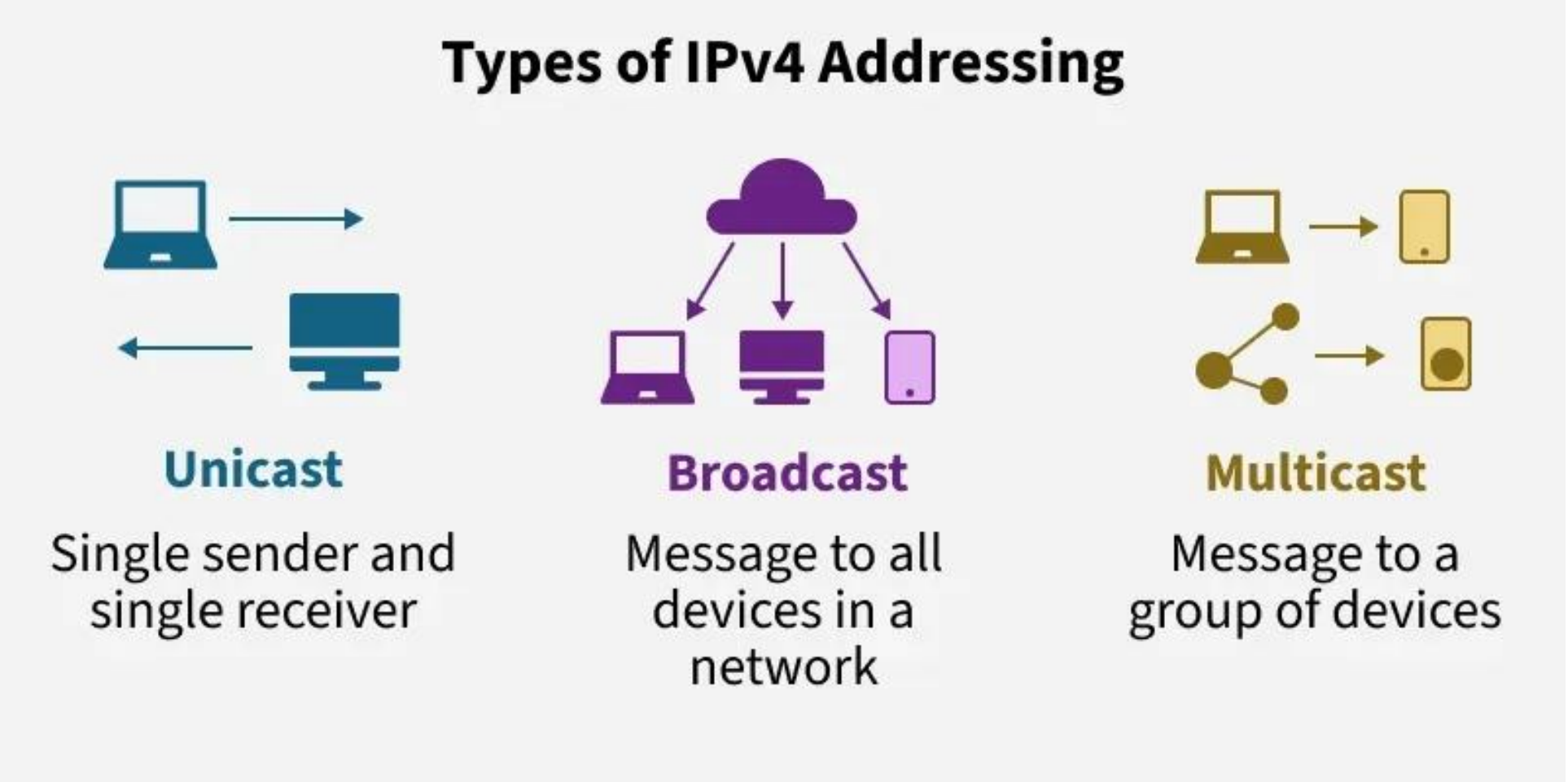
- **Large Address Space:** An IPv6 address is 128 bits long compared with the 32 bit address of IPv4, this is a huge(2 raised 96 times) increases in the address space.
- **Better Header Format**: IPv6 uses a new header format in which options are separated from the base header and inserted, when needed, between the base header and the upper layer data . This simplifies and speeds up the routing process because most of the options do not need to be checked by routers.
- **New Options**: IPv6 has new options to allow for additional functionalities.
- **Allowance for extension**: IPv6 is designed to allow the extension of the protocol if required by new technologies or applications.
- **Support For Resource Allocation**: In IPv6,the type of service field has been removed, but two new fields , traffic class and flow label have been added to enables the source to request special handling of the packet . this mechanism can be used to support traffic such as real-time audio and video.

- **Support For More Security:** The encryption and authentication options in IPv6 provide confidentiality and integrity of the packet. In IPv6 representation, we have three addressing methods :
  - Unicast
  - Multicast
  - Anycast

Addressing Methods

- **Unicast Address** : Unicast Address identifies a single network interface. A packet sent to a unicast address is delivered to the interface identified by that address.
- **Multicast Address** : Multicast Address is used by multiple hosts, called as **groups**, acquires a multicast destination address. These hosts need not be geographically together. If any packet is sent to this multicast address, it will be distributed to all interfaces corresponding to that multicast address. And every node is configured in the same way. In simple words, one data packet is sent to multiple destinations simultaneously.
- **Anycast Address*:*** Anycast Address is assigned to a group of interfaces. Any packet sent to an anycast address will be delivered to only one member interface (mostly nearest host possible).

Types of IPv4 Addressing



Internet Control Message Protocol (ICMP)

- Internet Control Message Protocol is known as ICMP.
- The protocol is at the network layer.
- It is mostly utilized on network equipment like routers and is utilized for error handling at the network layer.
- Since IP does not have an inbuilt mechanism for sending error and control messages. It depends on Internet Control Message Protocol(ICMP) to provide error control.
- It is a supporting protocol and is used by network devices like routers for sending error messages and operations information. For example, the requested service is not available or a host or router could not be reached.

Uses of ICMP

- ICMP is used for error reporting if two devices connect over the internet and some error occurs, So, the router sends an ICMP error message to the source informing about the error. For Example, whenever a device sends any message which is large enough for the receiver, in that case, the receiver will drop the message and reply to the ICMP message to the source.
- Another important use of ICMP protocol is used to perform network diagnosis by making use of traceroute and ping utility.

➢ **Traceroute:** Traceroute utility is used to know the route between two devices connected over the internet. It routes the journey from one router to another, and a traceroute is performed to check network issues before data transfer.

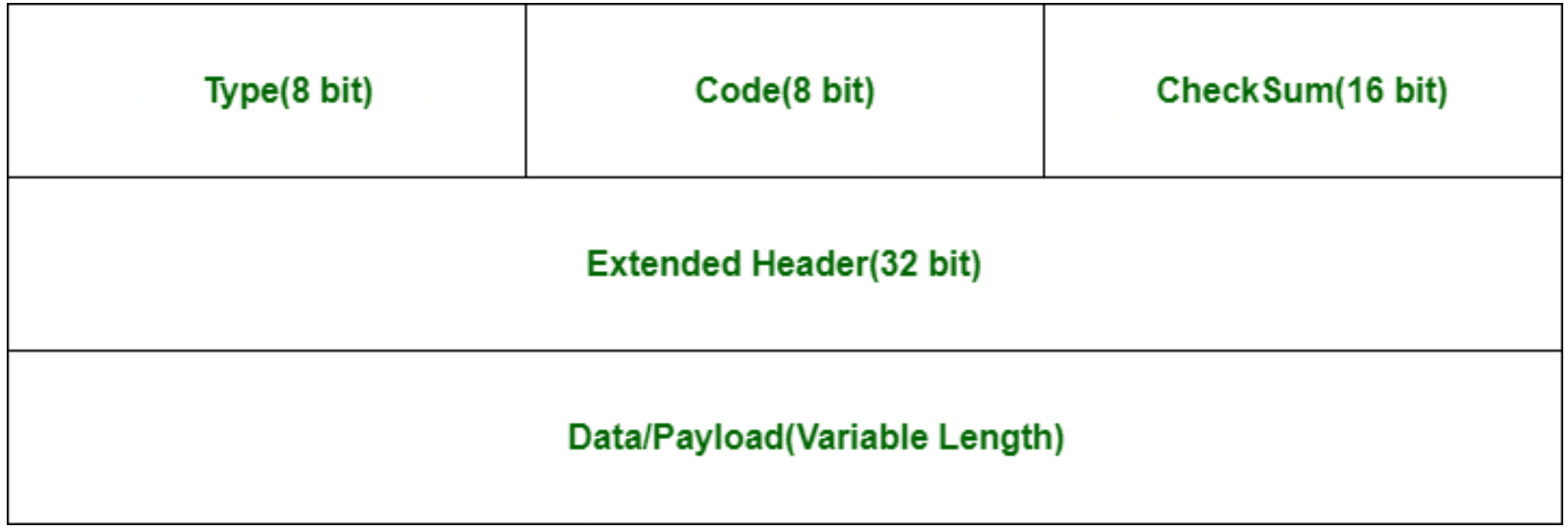
➢ **Ping:** Ping is a simple kind of traceroute known as the echo-request message, it is used to measure the time taken by data to reach the destination and return to the source, these replies are known as echo-replies messages.

Working of ICMP

- The working of ICMP is just contrasting with TCP, as TCP is a connection-oriented protocol whereas ICMP is a connectionless protocol.
- Whenever a connection is established before the message sending, both devices must be ready through a TCP handshake.
- ICMP packets are transmitted in the form of datagrams that contain an IP header with ICMP data.

ICMP Packet Format

ICMP header comes after IPv4 and IPv6 packet header.



In the ICMP packet format, the first 32 bits of the packet contain three fields:

- **Type (8-bit):** The initial 8-bit of the packet is for message type, it provides a brief description of the message so that receiving network would know what kind of message it is receiving and how to respond to it. Some common message types are as follows:
- Type 0 - Echo reply
- Type 3 - Destination unreachable
- Type 5 - Redirect Message
- Type 8 - Echo Request
- Type 11 - Time Exceeded
- Type 12 - Parameter problem
- **Code (8-bit):** Code is the next 8 bits of the ICMP packet format, this field carries some additional information about the error message and type.

Continued…

- **Checksum (16-bit):** Last 16 bits are for the checksum field in the ICMP packet header. The checksum is used to check the number of bits of the complete message and enable the ICMP tool to ensure that complete data is delivered.

- The next 32 bits of the ICMP Header are Extended Header which has the work of pointing out the problem in IP Message. Byte locations are identified by the pointer which causes the problem message and receiving device looks here for pointing to the problem.

- The last part of the ICMP packet is Data or Payload of variable length. The bytes included in IPv4 are 576 bytes and in IPv6, 1280 bytes.

Types of ICMP Messages

Type	Code	Description
0 - Echo Reply	0	Echo reply
3 - Destination Unreachable	0	Destination network unreachable
	1	Destination host unreachable
	2	Destination protocol unreachable
	3	Destination port unreachable
	4	Fragmentation is needed and the DF flag set
	5	Source route failed

Type	Code	Description
5 - Redirect Message	0	Redirect the datagram for the network
	1	Redirect datagram for the host
	2	Redirect the datagram for the Type of Service and Network
	3	Redirect datagram for the Service and Host
8 - Echo Request	0	Echo request
9 - Router Advertisement	0	Use to discover the addresses of operational routers
10 - Router Solicitation	0	
11 - Time Exceeded	0	Time to live exceeded in transit
	1	Fragment reassembly time exceeded.

Type	Code	Description
12 - Parameter Problem	0	The pointer indicates an error.
	1	Missing required option
	2	Bad length
13 - Timestamp	0	Used for time synchronization
14 - Timestamp Reply	0	Reply to Timestamp message

ICMP in DDoS attacks

- ICMP lacks authentication mechanisms and can be used by attackers for carrying out DoS and distributed DoS attacks.

- In DDoS attack, attackers overwhelm the target with unwanted traffic so the target can't provide service to its users.

There are multiple ways an attacker can use ICMP to execute these attacks, including the following:

1. Ping of death

- ✓ The attacker sends an IP packet larger than the number of bytes allowed by IP.
- ✓ On the way to its intended destination, the oversized packet is fragmented.
- ✓ However, when the recipient device reassembles it, the size exceeds the limit, causing a buffer overflow and the receiving machine to freeze or crash.

2. ICMP flood attack

- ✓ Sometimes called a *ping flood attack*, the goal of this attack is to overwhelm the target device with echo request packets.
- ✓ Each echo request packet must be processed by the target and responded to with Echo Reply messages.
- ✓ This sucks up all the target computer's resources, causing DoS to any other users of the target computer.

3. **Smurf attack.**

- ✓ In a Smurf attack, the attacker sends an ICMP packet with a spoofed source IP address, and the network layer equipment replies to the packet, sending the spoofed address a flood of packets. legacy equipment.

4. ICMP tunneling

- ✓ ICMP tunneling is a command-and-control attack technique in which attackers encapsulate and conceal malicious traffic within seemingly benign ICMP headers, making it bypass firewalls and network security measures.

5. ICMP router advertising spoofing

- ✓ During this attack, an attacker sends false ICMP router advertisement messages to a network, misleading devices into using the attacker's machine as their default gateway.
- ✓ This can lead to man- in-the-middle attacks, where the attacker intercepts and potentially alters communications between network devices.

Advantages of ICMP

- Network devices use ICMP to send error messages, and administrators can use the Ping and Tracert commands to debug the network.
- These alerts are used by administrators to identify issues with network connectivity.
- A prime example is when a destination or gateway host notifies the source host via an ICMP message if there is a problem or a change in network connectivity that needs to be reported. Examples include when a destination host or networking becomes unavailable, when a packet is lost during transmission, etc.
- Furthermore, network performance and connection monitoring tools commonly employ ICMP to identify the existence of issues that the network team has to resolve.
- One quick and simple method to test connections and find the source is to use the ICMP protocol, which consists of queries and answers.

Disadvantages of ICMP

- If the router drops a packet, it may be due to an error; but, because to the way the IP (internet protocol) is designed, there is no way for the sender to be notified of this problem.

- Assume, while a data packet is being transmitted over the internet, that its lifetime is over and that the value of the time to live field has dropped to zero. In this case, the data packet is destroyed.

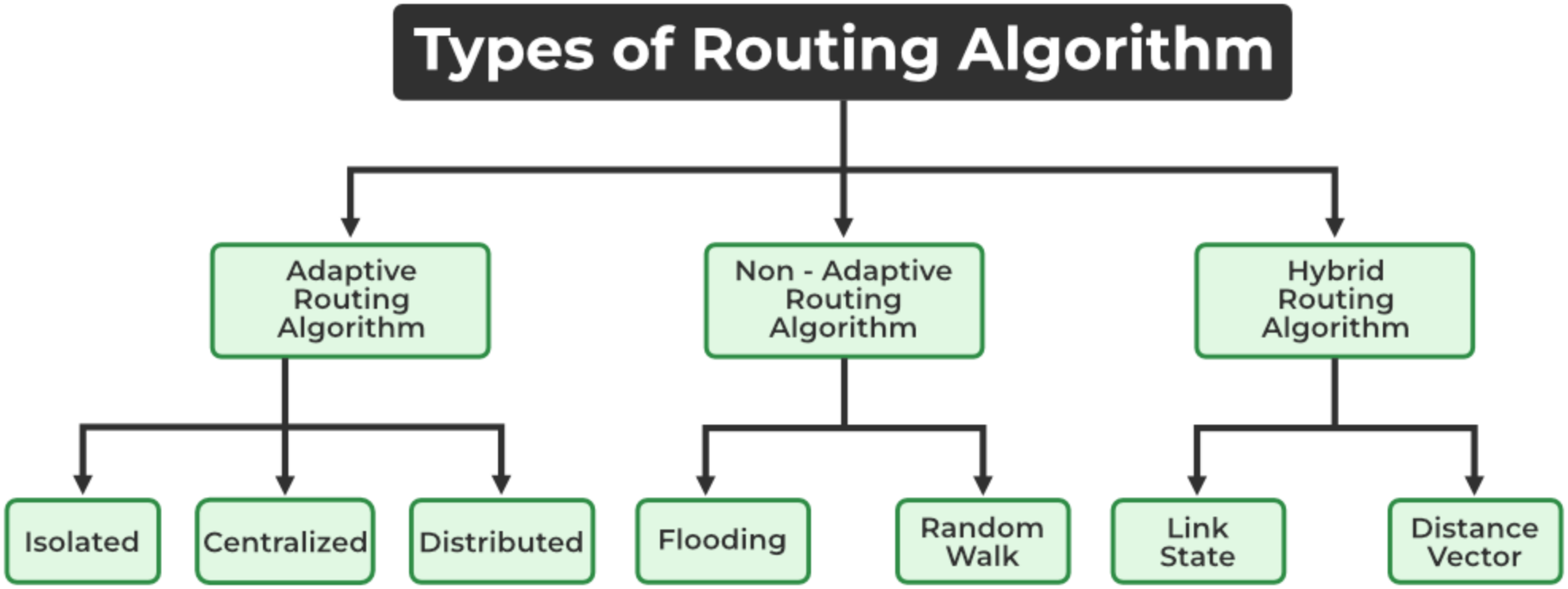
- Although devices frequently need to interact with one another, there isn't a standard method for them to do so in Internet Protocol. For instance, the host needs to verify the destination's vital signs to see if it is still operational before transmitting data.

Routing Algorithms

- A routing algorithm is a procedure that lays down the route or path to transfer data packets from source to the destination.
- They help in directing Internet traffic efficiently.
- After a data packet leaves its source, it can choose among the many different paths to reach its destination.
- Routing algorithm mathematically computes the best path, i.e. least cost path that the packet can be routed through.

Types of Routing Algorithms

Routing algorithms can be broadly categorized into two types, adaptive and nonadaptive routing algorithms.



1. Adaptive Algorithms

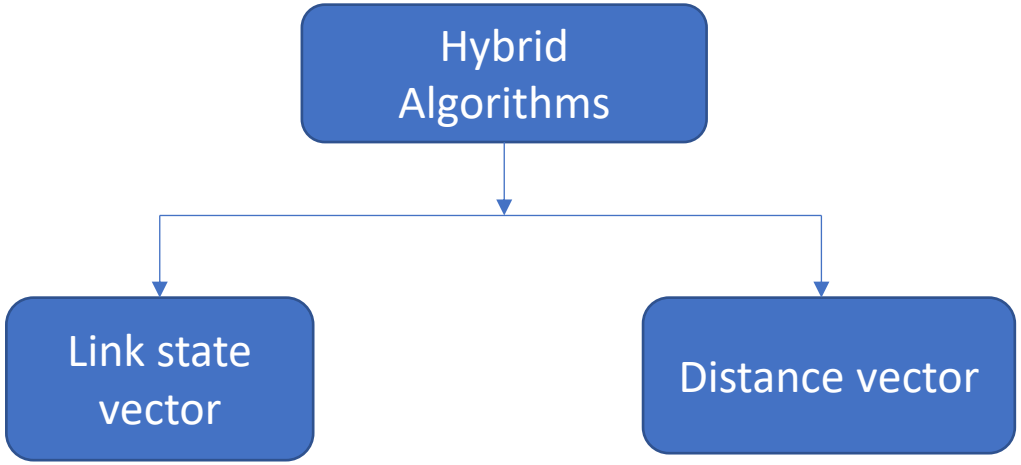
- These are the algorithms that change their routing decisions whenever network topology or traffic load changes.
- The changes in routing decisions are reflected in the topology as well as the traffic of the network.
- Also known as dynamic routing, these make use of dynamic information such as current topology, load, delay, etc. to select routes.

2. Non-Adaptive Algorithms

- These are the algorithms that do not change their routing decisions once they have been selected.
- This is also known as static routing as a route to be taken is computed in advance and downloaded to routers when a router is booted.

3. Hybrid Algorithms

- As the name suggests, these algorithms are a combination of both adaptive and non-adaptive algorithms.
- In this approach, the network is divided into several regions, and each region uses a different algorithm.



Link State Routing Algorithm

- A Link State Routing Algorithm is a type of routing algorithm used in computer networks to determine the shortest path from one node to another.
- Link state routing is a method in which each router shares its neighborhoods knowledge with every other router in the internetwork.
- In this algorithm, each router in the network understands the network topology then makes a routing table depend on this topology.
- It operates by constructing a complete topological map of the network, representing each node and link, and then using this map to calculate the shortest paths

Important Points Related to the Link State Routing Algorithm

- **Complete Topological Map:** It constructs a complete map of the network, including all nodes and links.
- **Dijkstra's Algorithm:** It often employs Dijkstra's algorithm to calculate the shortest paths between nodes.
- **Link State Packet:** Each router sends periodic Link State Packets (LSPs) containing information about its local links to all other routers in the network.
- **Topology Database:** Routers maintain a topology database containing information from received LSPs, allowing them to construct the network map.
- **Shortest Path Calculation:** After constructing the network map, routers use Dijkstra's algorithm to calculate the shortest paths to all other nodes.
- **Link State Updates:** When there are changes in the network topology, routers generate new LSPs to inform other routers, ensuring that all routers have up-to-date information.

Link State Routing Protocols

- Link state routing protocol maps the entire network, including the routers and the links between them.
- This map is called the link state database or topology table.
- The LSDB(Link State Database) is a very important component of a Link State Routing Protocol.
- It is a data structure that stores the topology of a network as known by a specific router running a Link State Routing Protocol.
- Link state routing protocols exchange LSAs(Link State Advertisements) between routers.
- It contains information about the state of the links connected to each router.
- The LSAs are used to build and maintain a complete and up-to-date network map.
- It enables routers to calculate the shortest path to a destination.

Phases of Link State Routing

There are two primary phases in link state routing:

1. **Flooding Phase**

- In this phase, each router floods its own LSA to all other routers in the network.
- The LSA contains information about the router's own links.
- It also contains the state of its neighboring routers.
- Routers use the LSA to build their own LSDB and to update the database as changes occur in the network.

2. **Calculation Phase**

- Once each router has received all LSAs from its neighbors and has built its own LSDB, it performs a shortest-path-first (SPF) calculation.
- It performs SPF to determine the best path to each destination in the network.
- The SPF calculation considers the cost of each link. It also considers the state of each router in the LSDB.
- The result of the SPF calculation is used to build the forwarding table. This table contains the best path to each destination.
- This table is used to forward packets to their intended destination.
- These two phases are continuous and occur simultaneously in the link state routing protocol.
- Routers continue to flood LSAs and update their LSDBs as changes occur in the network. They recalculate the SPF algorithm to determine the best path to each destination based on the most up-to-date network topology information.

Algorithm for Link State Routing

1.Discovery phase:

- ✓Each router sends out Hello packets to discover its neighbors and to establish a neighbor adjacency.
- ✓Once a neighbor adjacency is established, routers exchange LSAs to learn about the state of the network.

2.LSA flooding:

- ✓Each router floods its own LSA to all other routers in the network.
- ✓ The LSA consists of information about the router's own links and the state of its neighboring routers.
- ✓Routers use the LSA to build their own LSDB and to update the database as changes occur in the network.

3. **Shortest Path First (SPF) calculation:**

- ✓Once each router has received all LSAs from its neighbors and has built its own LSDB, it performs an SPF calculation which determines the best path to each destination in the network.
- ✓The SPF calculation considers the cost of each link and the state of each router in the LSDB.
- ✓The result of the SPF calculation is used to build the forwarding table.
- ✓This table contains the best path to each destination. This table is used to forward packets to their intended destination.

4.Updating LSAs:

- ✓When a change occurs in the network, such as a link failure or adding a new router, the affected router floods a new LSA to all other routers in the network.
- ✓ This triggers a recalculation of the SPF algorithm, and the forwarding table is updated to reflect the new best path to each destination.

5. Aging LSAs:

- ✓ To prevent outdated LSAs from remaining in the LSDB, each router assigns a time-to-live (TTL) value to each LSA.
- ✓ When the TTL value expires, the router removes the LSA from its LSDB.

The above steps are repeated continuously to maintain an up-to-date view of the network topology and to determine the best path to each destination.

Advantages of Link State Routing

- Since each router has a complete network map, link state routing protocols can converge much faster than distance vector protocols, which rely on periodic updates.

- Link state routing protocols only send updates when there is a change in the network topology. It reduces the amount of bandwidth used compared to distance vector protocols.

- Link state routing protocols are better suited to large networks. They can handle a more significant number of routers and links without suffering from routing loops or other issues.

Disadvantages of Link State Routing

- Link state routing protocols require more computational power. It requires more memory to maintain the link state database and to perform the shortest path calculation. This can limit their scalability, especially in more extensive networks.
- The link state routing protocol is more complex than other routing protocols. This complexity makes the protocol more challenging to configure and manage.
- It generates more network traffic than distance vector protocols. It sends frequent updates about changes in the network topology.
- These protocols are vulnerable to routing attacks, such as spoofed LSA attacks. It can cause routers to make incorrect routing decisions.

Routing Information Protocol

- Routing Information Protocol (RIP) is a distance vector protocol that uses hop count as its primary metric.
- RIP defines how routers should share information when moving traffic among an interconnected group of local area networks.
- RIP was originally designed for the Xerox PARC Universal Packet protocol suite and was known as Gateway Information Protocol in the Xerox Network Systems protocol suite in 1981.
- RIP -- defined in Request for Comments 1058 in 1988 -- is easy to configure and easy to use in small networks.

How RIP works

- RIP uses a distance vector algorithm to decide which path to put a packet on so it arrives at its destination.
- Each RIP router maintains a routing table listing all the destinations the router knows how to reach and broadcasts it to its closest neighbors every 30 seconds.
- In this context, *neighbors* are the other routers that the first router connects directly to.
- This only includes the other routers on the same network segments as the selected router.
- The neighbors, in turn, pass the information on to their nearest neighbors, and so on, until all RIP hosts within the network have the same knowledge of routing paths. This shared knowledge is known as *convergence*.

Continued…

- If a router receives an update on a route and the new path is shorter, it updates its table entry with the shorter path's length and next-hop address.
- If the new path is longer, it waits through a hold-down period to see if later updates also reflect the higher value.
- It only updates the table entry if it determines the new, longer path is stable.
- If a router crashes or something severs the network connection, the router stops sending and receiving updates to or from its neighbors or along the severed connection.
- If a given route in the routing table isn't updated across six successive update cycles -- 180 seconds -- a RIP router drops that route and notifies the rest of the network about the problem through its own periodic updates.

Hop counts

- RIP only supports 15 hops in a path.
- By default, a router has one hop if its neighbor owns a destination network and can deliver packets directly without using other routers.
- Network management terminology describes this as a *cost of one*.
- If a packet can't reach a destination in 15 hops, it's considered unreachable.
- RIP uses a modified hop count to determine network distance. A modified hop count reflects how network engineers can assign paths a higher cost -- as if they involved extra hops -- if they want to limit or discourage their use. For example, they might assign a satellite backup link a cost of 10 hops to force traffic to follow other routes when available.

RIP protocol runs with an Administrative Distance (AD) of 120 and port 520 on the OSI model of network layer

- **Command** It is used for request and response. It is an 8-bit field. The request is represented by 1 whereas the response is represented by 2
- **Version** Version represents the protocol version of RIP in use. Lets assume we use version 1 of the protocol, then we enter 1 in this field
- **Reserved** It is a reserved filed and it's represented by 0
- **Family** This is a 16-bit field. Since we use TCP/IP protocol, we put 2 in this field
- **Network** Address: It's a 14 byte field. Since we use IPv4 which gets filled in 4 bytes, rest 10 bytes remain zero
- **Distance** The hope count or the total number of routers require to reach the destination network is represented by this field

RIP timers

RIP timers help regulate performance. They include the following:

- **Update timer**

✓This is the frequency of routing updates.

✓ Every 30 seconds, IP RIP sends a complete copy of its routing table, subject to split horizon. Internetwork Packet Exchange RIP does this every 60 seconds.

- **Invalid timer.**

✓ This is the absence of refreshed content in a routing update.

✓RIP waits 180 seconds to mark a route as invalid and immediately puts it into hold-down.

- **Hold-down timers and triggered updates**

✓This assists with route stability in a Cisco environment.

✓ Hold-downs ensure that regular update messages do not inappropriately cause a routing loop.

✓The router doesn't act on nonsuperior new information for a specific period of time.

✓ RIP's hold-down time is 180 seconds.

- **Flush timer**

✓RIP waits an additional 240 seconds after hold-down before it removes the route from the table.

Versions of RIP

There are in total 3 versions of RIP:

- RIPv1
- RIPv2
- RIPng

RIPv1

- Standardized in 1988, RIPv1 is also known as a *classful routing protocol* because it doesn't send subnet mask information in its routing updates.
- RIPv1 determines routes based on the Internet Protocol (IP) destination and hop count.
- The routing table broadcasts to every station on the attached network.

RIPv2

- Standardized in 1998, RIPv2 is also a classless routing protocol because it sends subnet mask information in its routing updates.
- RIPv2 advanced RIPv1's routing method and included subnet masks and gateways.
- RIPv2 sends the routing table to a multicast address to reduce network traffic.
- Additionally, RIPv2 uses authentication for security -- a feature missing from RIPv1.

RIPng

- RIPng is an extension of RIPv2 made to support IPv6, as the previous versions only ran in IPv4.

RIP Ver1	RIP Ver2
RIP v1 uses what is known as classful routing	RIP v2 is a classless protocol and it supports variable-length subnet masking (VLSM), CIDR, and route summarization
RIPv1 routing updates are broadcasted	RIP v2 routing updates are multicasted
RIPv1 has no authentication	RIP v2 supports authentication
RIP v1 does not carry mask in updates	RIP v2 does carry mask in updates, so it supports for VLSM
RIP v1 is an older, no longer much used routing protocol	IP v2 can be useful in small, flat networks or at the edge of larger networks because of its simplicity in configuration and usage

Open Shortest Path First (OSPF)

- \Open Shortest Path First (OSPF) is a link-state routing protocol used to find the best path between routers within an Autonomous System (AS).
- OSPF routers establish neighbor relationships to exchange routing information, and these relationships play a crucial role in the formation of OSPF adjacencies.
- Understanding the transitions between OSPF protocol states is essential for optimizing network performance and troubleshooting.
- OSPF uses Hello packets for neighbor discovery and to establish communication between routers.
- Neighbor states are key in forming OSPF adjacencies, enabling efficient routing.

Open Shortest Path First (OSPF) Terms

- **Router ID:** The Router ID is the highest active IP address on the router. The process first considers the highest loopback address. If no loopback address is configured, the highest active IP address on the router's interfaces is used as the Router ID.
- **Router Priority:** The router priority is an 8-bit value assigned to a router running OSPF, which is used during the election of the Designated Router(DR) and Backup Designated Router(BDR) in a broadcast network.
- **Designated Router (DR):** The Designated Router (DR) is elected to minimize the number of OSPF adjacencies in a network. It is responsible for distributing Link State Advertisements(LSAs) to all other routers. In a broadcast network, the DR handles LSR (Link State Request) messages and responds with updates.
- **Backup Designated Router (BDR):** The Backup Designated Router (BDR) serves as a backup to the DR in a broadcast network. If the DR fails, the BDR assumes the role of DR and takes over its responsibilities.

DR and BDR Election

The election of the DR and BDR occurs in broadcast or multi-access networks. The following criteria are used for the election:

1. The router with the highest priority is elected as the DR.

2. If there is a tie in the router priority, the router with the highest Router ID is chosen. The Router ID is first determined by the highest loopback address, and if no loopback is configured, the highest active IP address on the router’s interfaces is considered.

Open Shortest Path First (OSPF) States

The device operating OSPF goes through certain states. These states are:

Down State

- No Hello packets have been received on the interface.
- This is the initial state before communication begins between routers. The OSPF adjacency process has not started yet.

Init State

- The router receives a Hello packet from another router but is not yet listed in the neighbor’s Hello packet.
- The router has begun communication, but no bidirectional communication has been established yet.

Two-Way State

- Bi-directional communication is confirmed when each router sees itself in the other's Hello packet.
- In broadcast or multi-access networks, this is the point at which the Designated Router (DR) and Backup Designated Router (BDR) election takes place.

Continued…

ExStart State

- The Master/Slave relationship is established for Database Description (DBD) exchange.
- Routers exchange initial DBD packets, which contain the sequence numbers, indicating the beginning of the exchange phase. The router with the higher Router ID becomes the master, while the other becomes the slave.

Exchange State

- In this state, routers exchange full Database Description (DBD) packets, which contain the headers of Link State Advertisements (LSAs).
- Routers create Link-State Request (LSR) lists to request missing LSAs that are not present in their own Link-State Database (LSDB).

Loading State

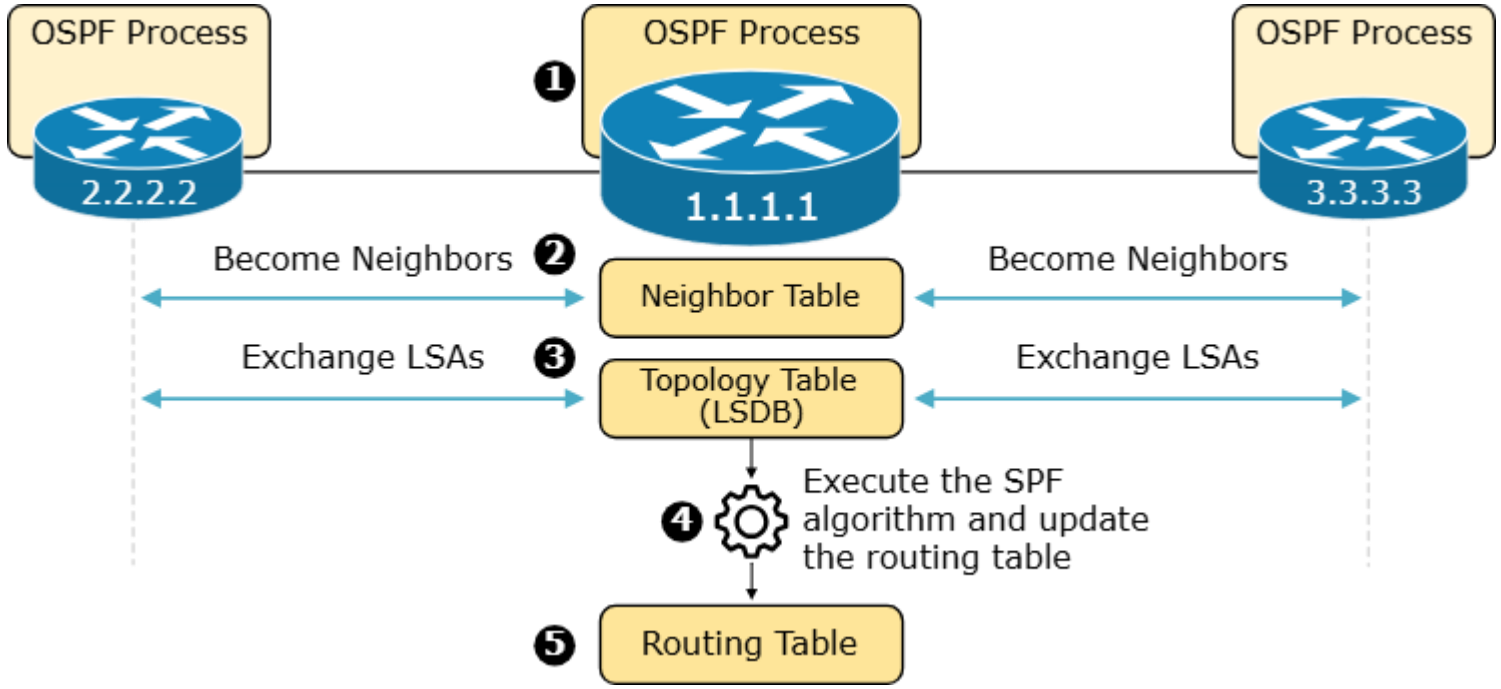
- Routers send Link-State Request (LSR) packets to request missing LSAs.
- The routers respond with Link-State Update (LSU) packets, which contain the requested LSAs.

Full State

- The adjacency between routers is fully established.
- The Link-State Databases (LSDB) are synchronized, meaning both routers have identical databases.
- Routers can now compute the shortest paths using the Shortest Path First (SPF) algorithm, marking the point where routing decisions can be made.

How does OSPF work?

To explain how the link-state protocol works, **we have broken down the OSPF operation into five steps**



Step 1. Enable the local OSPF process and choose RID

- The first step in the OSPF operation is to enable the local process on each router.
- When the process initializes, it must allocate a unique Router ID (RID) to be able to send OSPF messages.

Step 2. Establish neighbor adjacencies:

- Once the routing process is enabled on a router, it starts sending Hello messages on all OSPF-enabled links to determine whether neighbors are present on those segments.
- If other OSPF-enabled devices sit on the same segment, the router attempts to establish a neighbor adjacency with that device.

Step 3. Exchange LSAs and build the topology table (the LSDB)

- After the router becomes adjacent to the neighboring OSPF devices, it starts exchanging LSAs.
- Link-state Advertisements (LSAs) contain each directly connected link's cost and IP settings.
- Every router floods all LSAs to its neighbors until every device in the OSPF domain has all LSAs.
- This process is referred to as the LSA-flooding.
- The router builds the link-state database (LSDB) based on the received LSAs.
- The LSDB holds all the information about the network's topology (hence, it is called the topology table). Because all LSAs are flooded to every router in the area, all routers in the same area have the same information in their LSDBs.

Step 4. Execute the SPF algorithm

- The LSDB serves as an input to the SPF algorithm.
- The algorithm calculates the best paths based on the advertised cost of each link and creates the SPF tree from the point of view of the local router.

Step 5. Update the routing table with the best paths

- Lastly, the router adds the best paths from the SPF tree into the routing table.
- The router then makes forwarding decisions based on the entries in the routing table.

BGP

- Border Gateway Protocol (BGP) is a set of rules that determine the best network routes for data transmission on the internet.
- The internet consists of thousands of private, public, corporate, and government networks linked together through standardized protocols, devices, and communication technologies.
- When you browse the internet, data travels across multiple networks before reaching its destination.
- BGP's responsibility is to look at all the available paths that data could travel and select the best route.

Why is Border Gateway Protocol important?

- Border Gateway Protocol (BGP) makes the internet work through data routing.
- BGP routing is critical because, at its core, the internet is made of hundreds of thousands of autonomous systems.
- An autonomous system is a smaller network under the control of a single administrative entity.
- You can uniquely identify such networks by their autonomous system number assigned by the Internet Assigned Numbers Authority (IANA).
- Data travels between autonomous systems as it moves from source to destination.

BGP supports every autonomous system to do the following.

1. Find the best route

- As data travels across the internet from source to destination, every autonomous system in between has to decide where the data packet should go next.
- The decision is based on several factors like geographical location, network congestion, and data transfer cost.
- BGP routing considers these factors and helps determine the next best autonomous system so that data travels on the shortest route from source to destination

2. Discover network connection changes

- The structure of the internet is dynamic. New autonomous systems are being added, and old ones are being removed constantly.
- Every autonomous system must stay updated with information regarding new and obsolete routes. BGP helps systems to discover and remain updated on such network changes.

Continued…

3. Administer network policies

- BGP has the flexibility to allow autonomous system administrators to implement their own routing policies.
- For example, you can configure a router running BGP to distinguish between the routes that are internal and external to the autonomous system. The administrator can set rules to determine whether data should be routed internally or externally.

4. Add a layer of network security

- BGP supports security in your network management.
- For example, BGP can authenticate messages between routers using preconfigured passwords. Administrators can verify BGP messages that come from legitimate autonomous systems and filter out unauthorized traffic.

How does the Border Gateway Protocol work?

Border Gateway Protocol (BGP) works using a mechanism called *peering*. Administrators assign certain routers as BGP peer or BGP speaker routers.

BGP peers perform the following main functions.

1. Route discovery

- BGP peers exchange routing information with neighboring BGP peers through network-layer reachability information (NLRI) and path attributes.
- NLRI includes connectivity information about neighbors.
- Path attributes include information like latency, hop count, and cost of transmission.
- After they exchange information, each BGP peer can then construct a graph of network connections around it.

2. Route storage

- ✓ During the discovery process, every BGP router collects route advertisement information and stores it in the form of routing tables.
- ✓ It uses the routing table for path selection and also updates it frequently.
- ✓ For instance, the BGP router receives *keep-alive* messages every 30 seconds from neighboring routers. It updates the stored routes accordingly.

3. Path selection

- ✓ BGP routers use the stored information to route traffic optimally.
- ✓ The primary factor in path selection is the shortest path, as determined by the stored route graphs.
- ✓ When a destination is reachable from multiple paths, BGP selects the best one by sequentially evaluating other path attributes.

Types of Border Gateway Protocol

- Border Gateway Protocol (BGP) is classified as internal and external, depending on where the data is being routed.

- If a BGP session is established between two neighbors in different autonomous systems, the session is *external BGP (EBGP)*, and if the session is established between two neighbors in the same AS, the session is *internal BGP*

Difference Between BGP and OSPF

BGP	OSPF
It follows the Path Vector Routing Algorithm	It follows the Link State Routing Algorithm
The speed of convergence is very slow in BGP	The speed of convergence is fast in the case of OSPF
BGP is also called inter-domain routing protocol	OSPF is also called intra-domain routing protocol
In BGP routing operation is performed between two AS	In OSPF routing operation is performed inside an AS
In BGP, TCP protocol is used	In OSPF, IP protocol is used

Figure illustrates this concept.

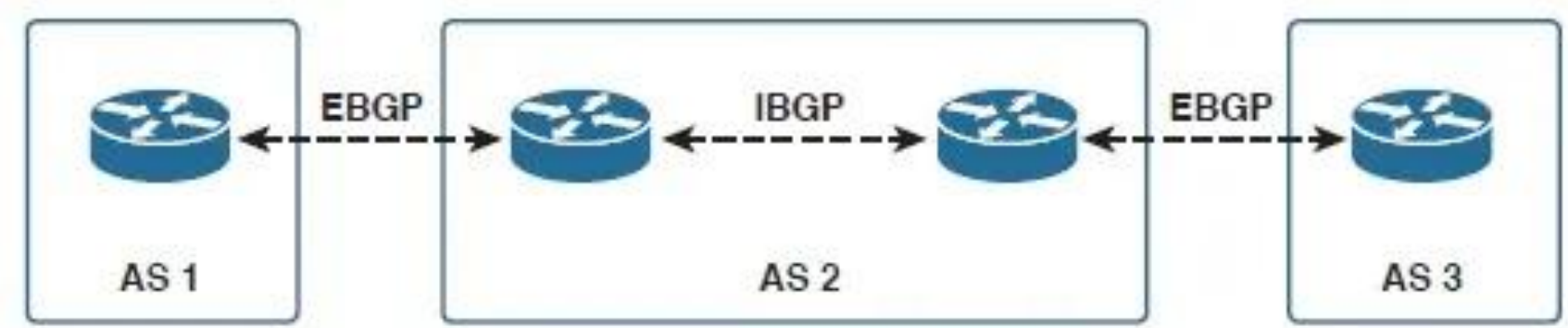


Figure 1-6 *Neighbors in Different Autonomous Systems Communicate Using EBGP, Whereas Neighbors in the Same AS Communicate Using IBGP*